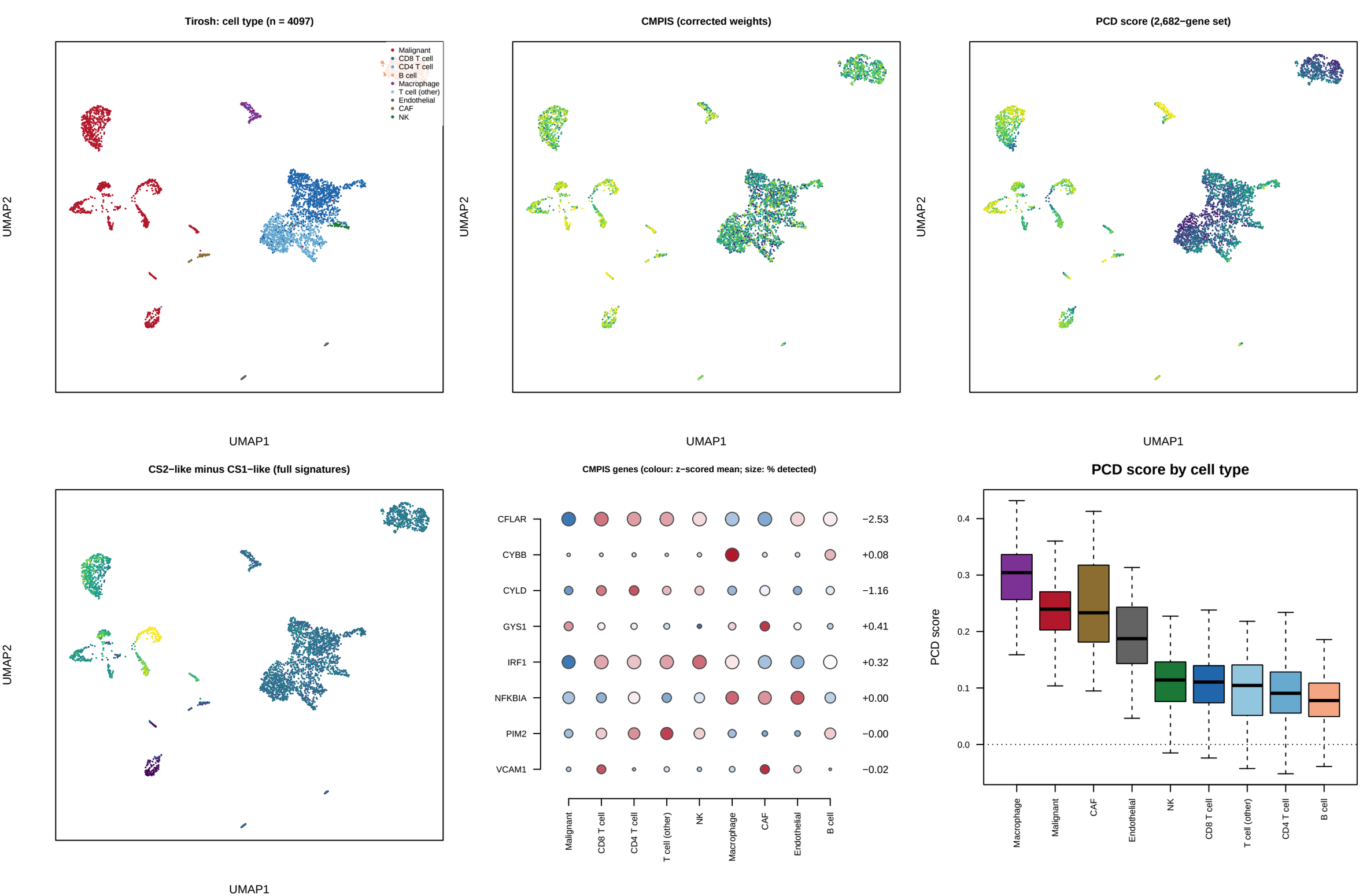
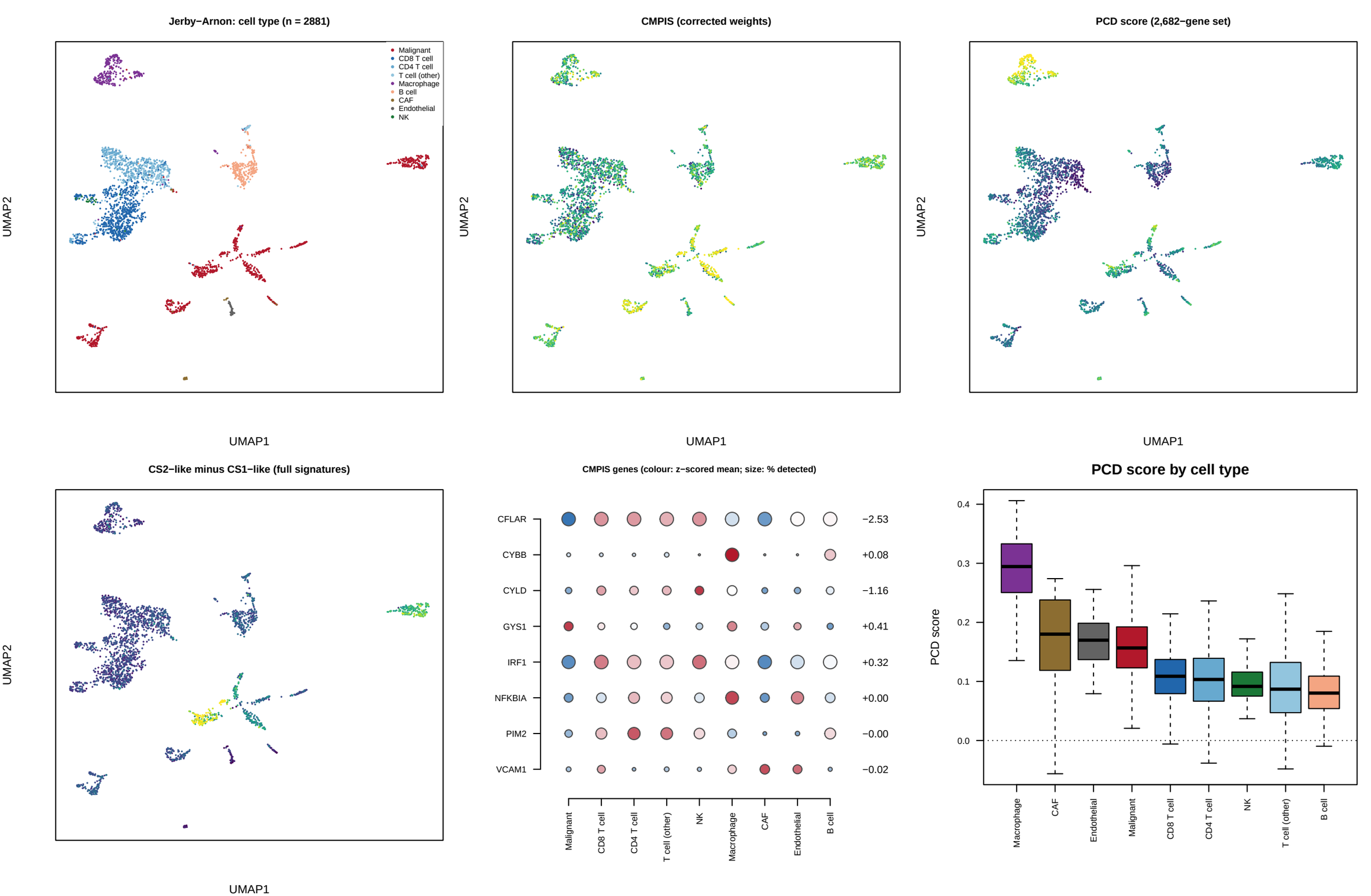
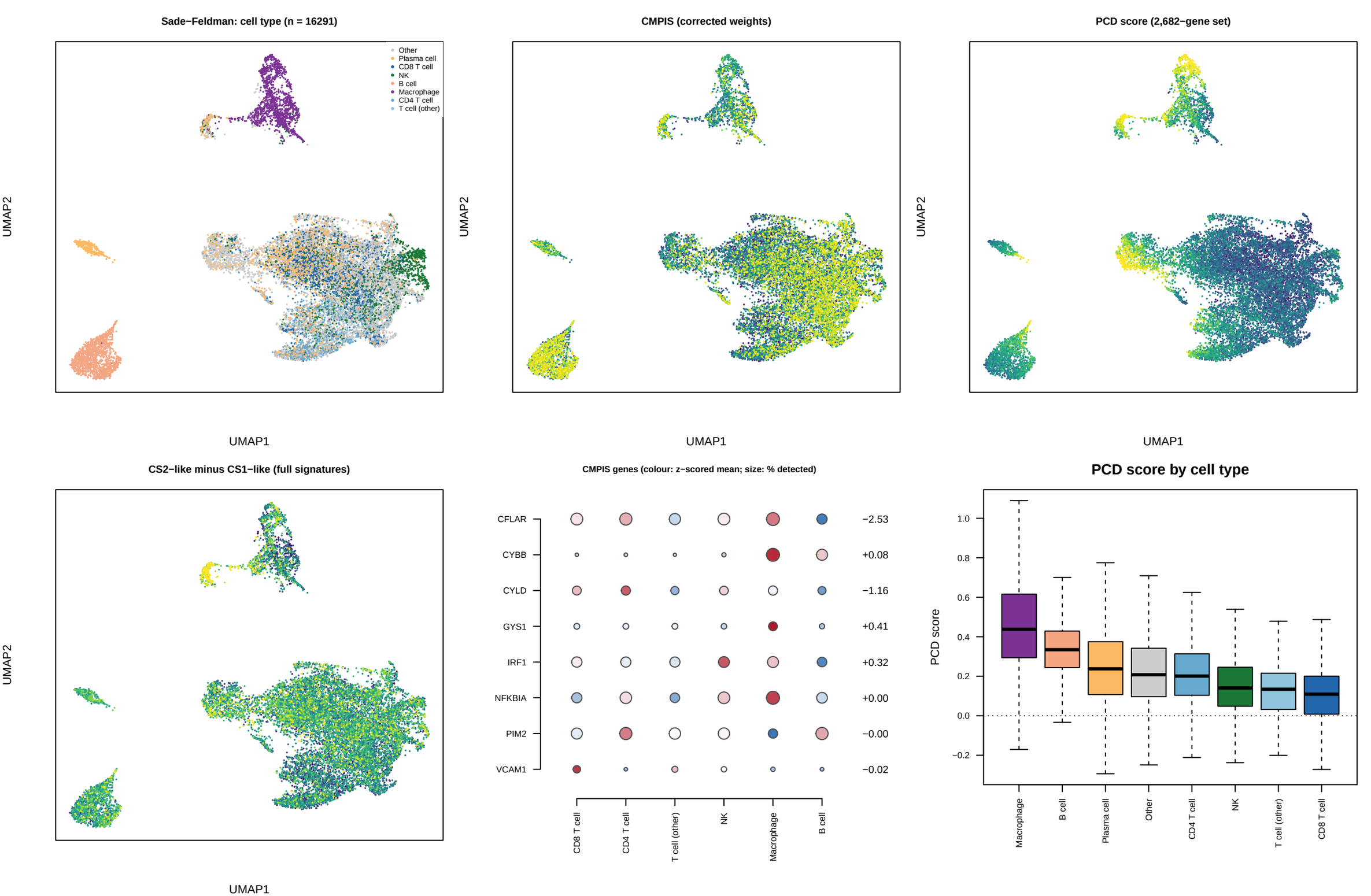
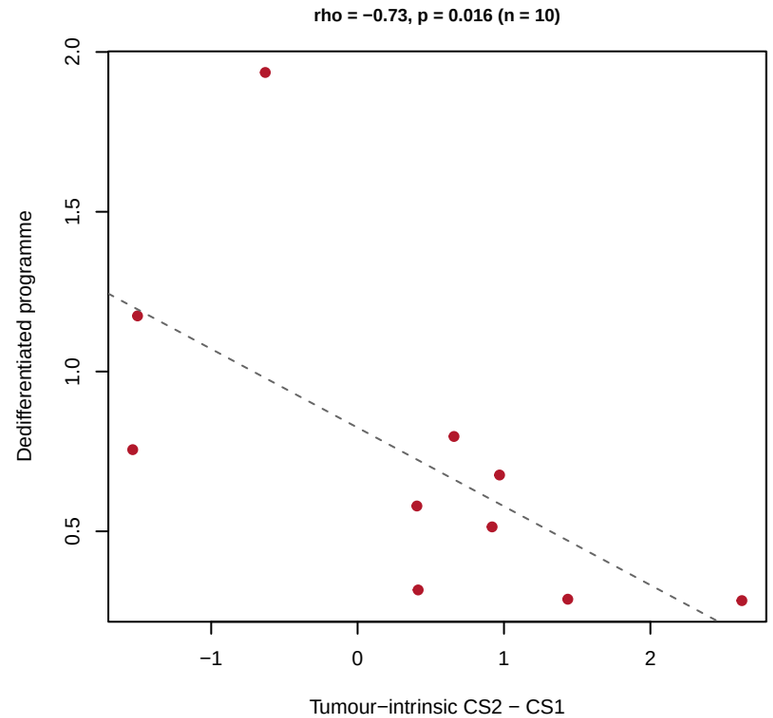
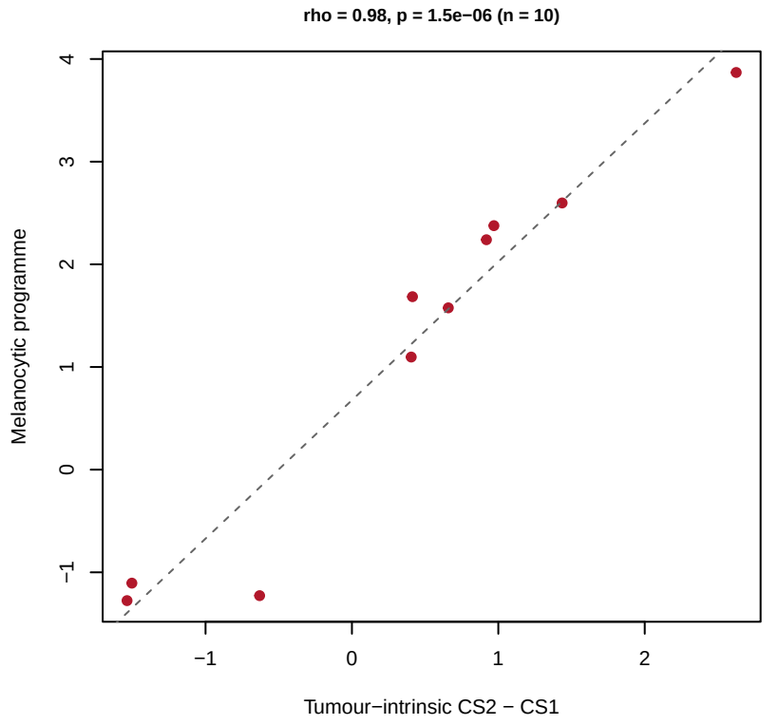

Tirosh: cell type (n = 4097)
Malignant
CD8 T cell
CD4 T cell
B cell
Macrophage
T cell (other)
Endothelial
CAF
NK
UMAP2
UMAP1
CMPIS (corrected weights)
UMAP2
UMAP1
PCD score (2,682−gene set)
UMAP2
UMAP1
CS2−like minus CS1−like (full signatures)
UMAP2
UMAP1
CMPIS genes (colour: z−scored mean; size: % detected)
CFLAR
CYBB
CYLD
GYS1
IRF1
NFKBIA
PIM2
VCAM1
−2.53
+0.08
−1.16
+0.41
+0.32
+0.00
−0.00
−0.02
Malignant
CD8 T cell
CD4 T cell
T cell (other)
NK
Macrophage
CAF
Endothelial
B cell
PCD score by cell type
PCD score
0.0
0.1
0.2
0.3
0.4
Macrophage
Malignant
CAF
Endothelial
NK
CD8 T cell
T cell (other)
CD4 T cell
B cell

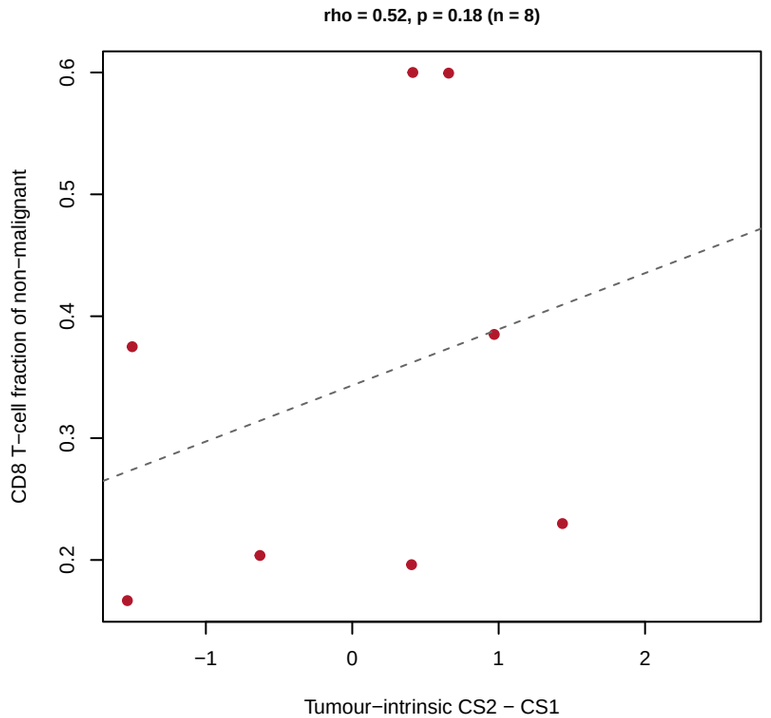

Jerby−Arnon: cell type (n = 2881)
Malignant
CD8 T cell
CD4 T cell
T cell (other)
Macrophage
B cell
CAF
Endothelial
NK
UMAP2
UMAP1
CMPIS (corrected weights)
UMAP2
UMAP1
PCD score (2,682−gene set)
UMAP2
UMAP1
CS2−like minus CS1−like (full signatures)
UMAP2
UMAP1
CMPIS genes (colour: z−scored mean; size: % detected)
CFLAR
CYBB
CYLD
GYS1
IRF1
NFKBIA
PIM2
VCAM1
−2.53
+0.08
−1.16
+0.41
+0.32
+0.00
−0.00
−0.02
Malignant
CD8 T cell
CD4 T cell
T cell (other)
NK
Macrophage
CAF
Endothelial
B cell
PCD score by cell type
PCD score
0.4
0.3
0.2
0.1
0.0
Macrophage
CAF
Endothelial
Malignant
CD8 T cell
CD4 T cell
NK
T cell (other)
B cell

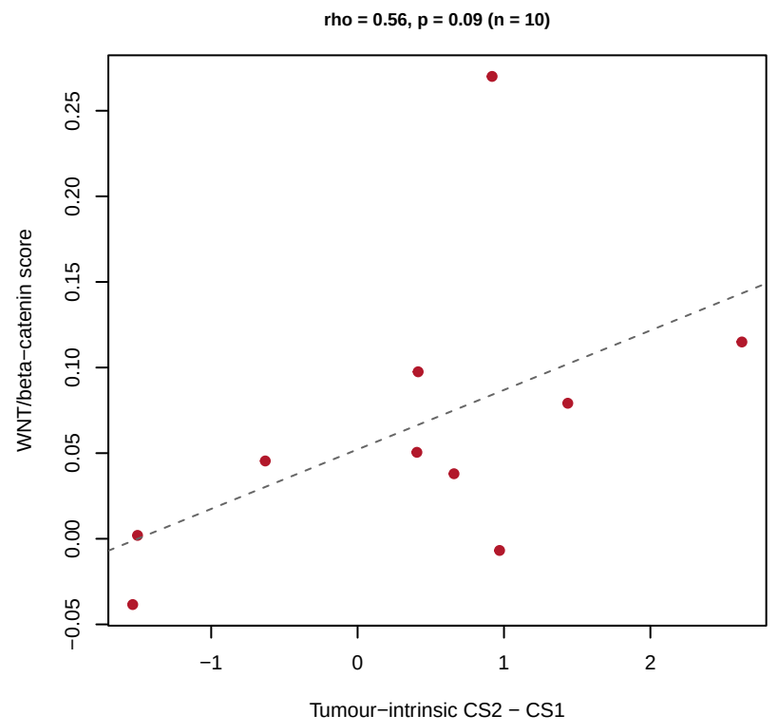

Sade−Feldman: cell type (n = 16291)
Other
Plasma cell
CD8 T cell
NK
B cell
Macrophage
CD4 T cell
T cell (other)
UMAP2
UMAP1
CMPIS (corrected weights)
UMAP2
UMAP1
PCD score (2,682−gene set)
UMAP2
UMAP1
CS2−like minus CS1−like (full signatures)
UMAP2
UMAP1
CMPIS genes (colour: z−scored mean; size: % detected)
CFLAR
CYBB
CYLD
GYS1
IRF1
NFKBIA
PIM2
VCAM1
−2.53
+0.08
−1.16
+0.41
+0.32
+0.00
−0.00
−0.02
CD8 T cell
CD4 T cell
T cell (other)
NK
Macrophage
B cell
PCD score by cell type
PCD score
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Macrophage
B cell
Plasma cell
Other
CD4 T cell
NK
T cell (other)
CD8 T cell

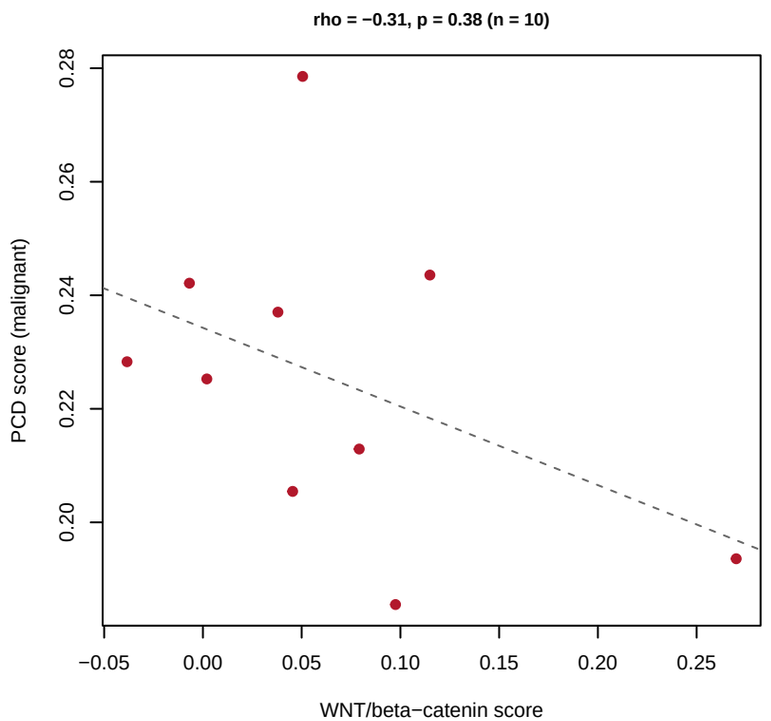
Tirosh: one point = one tumour (>= 15 malignant cells)

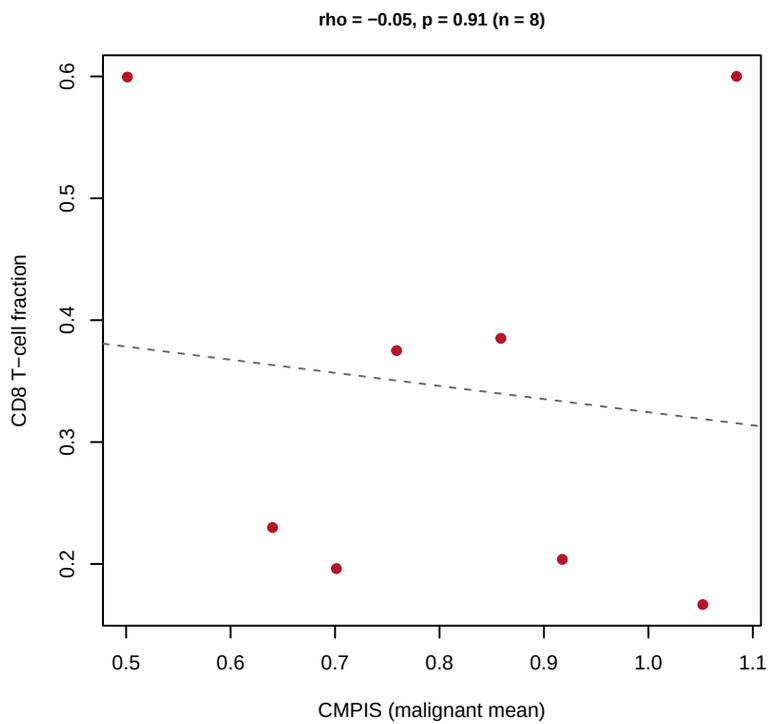


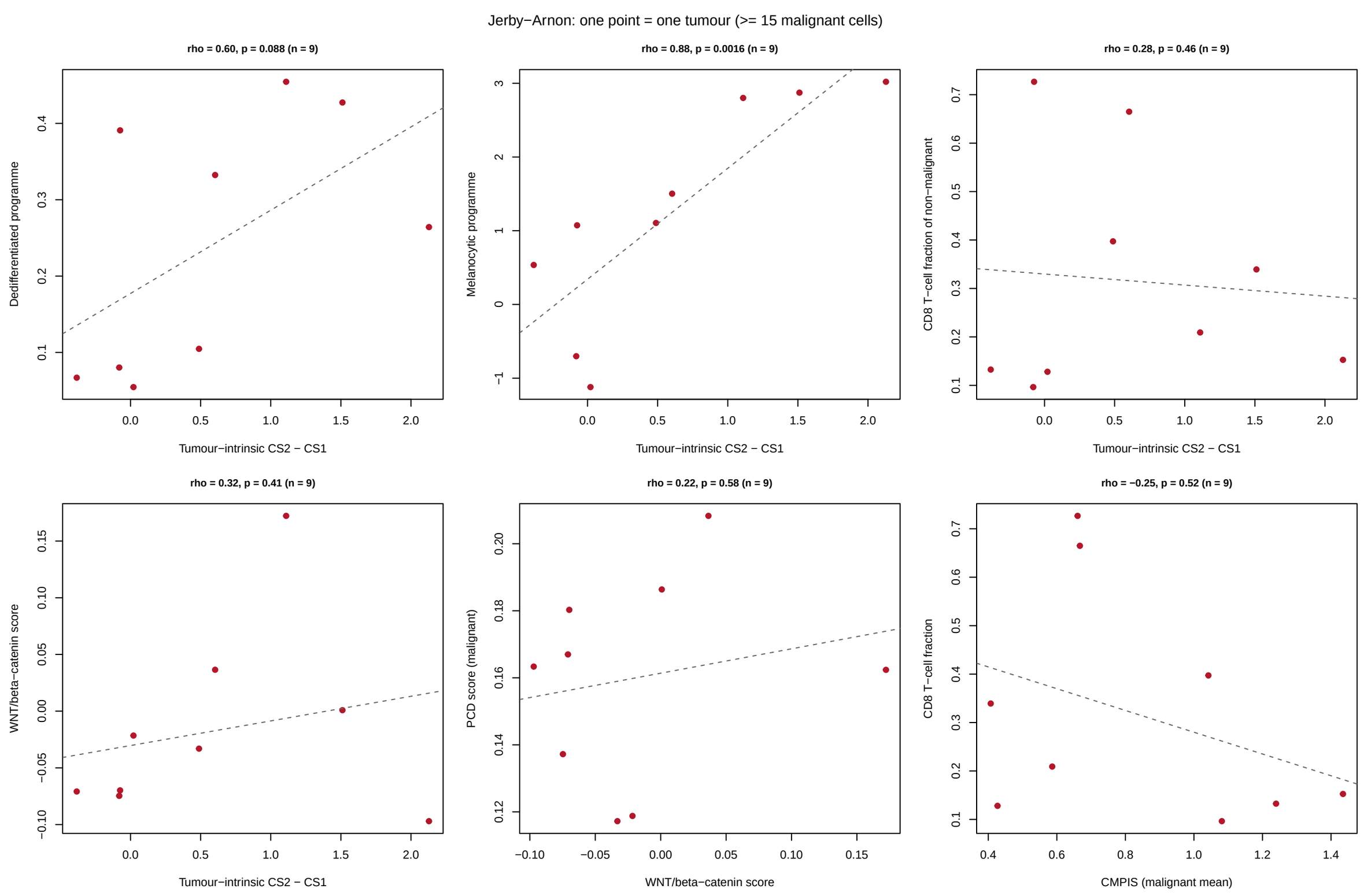


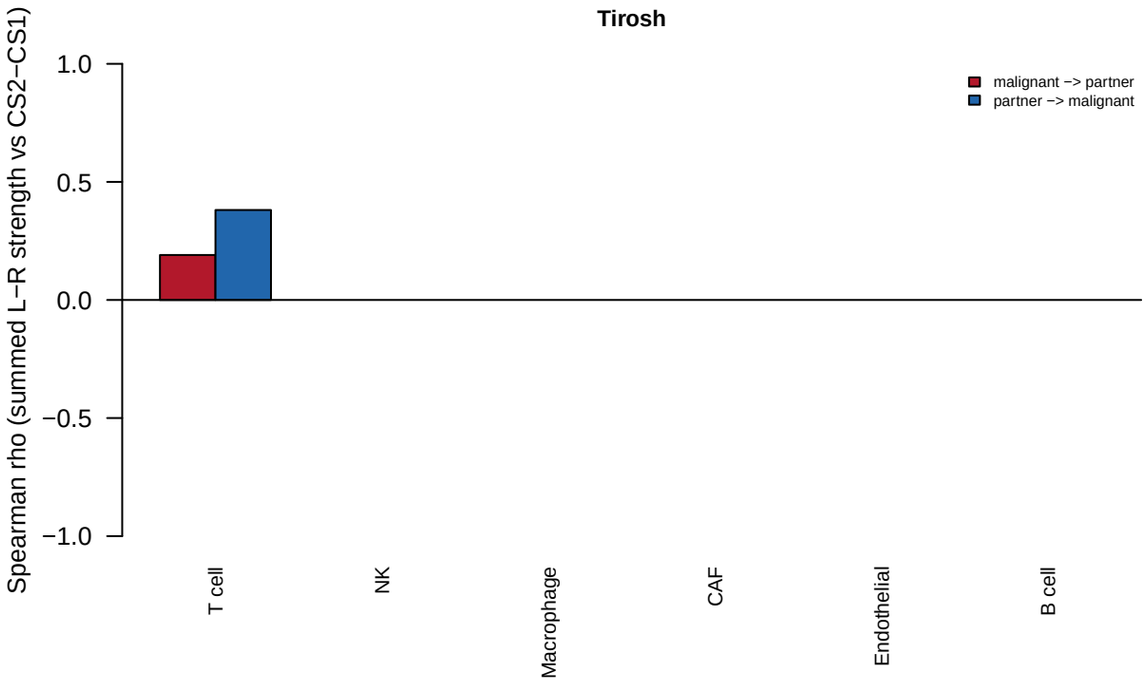


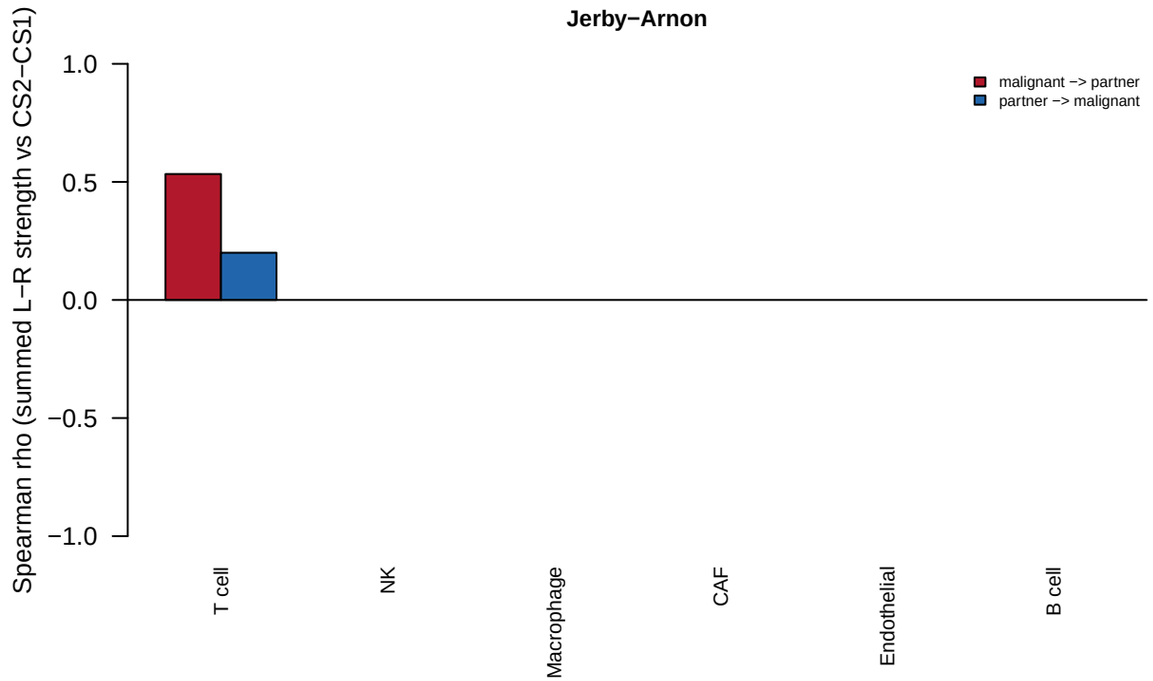


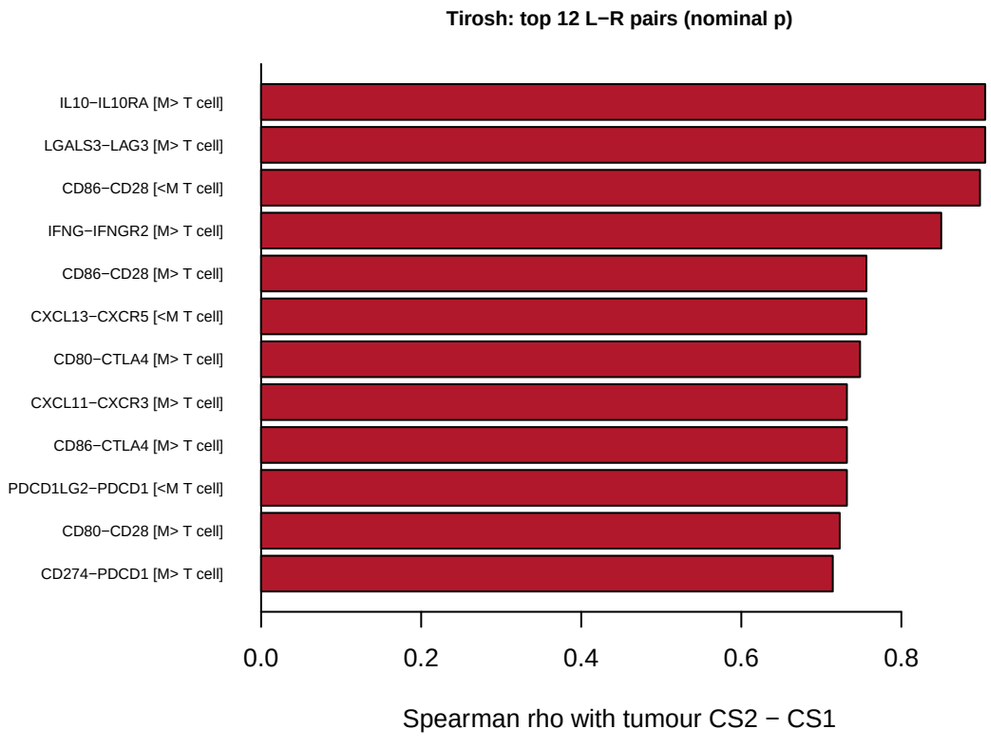


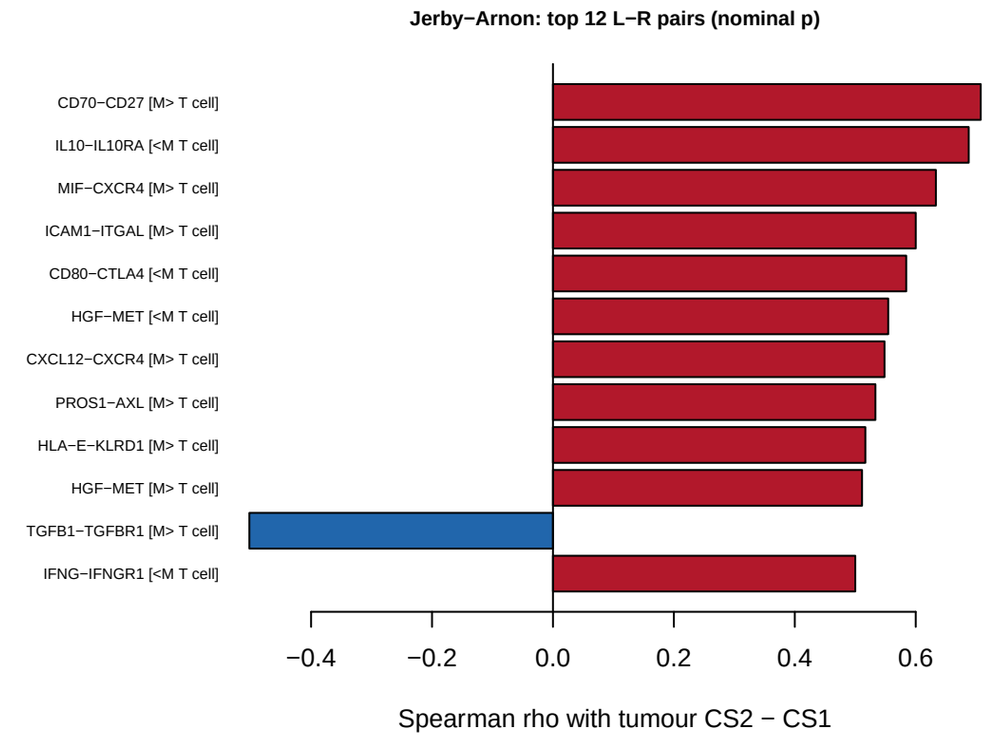


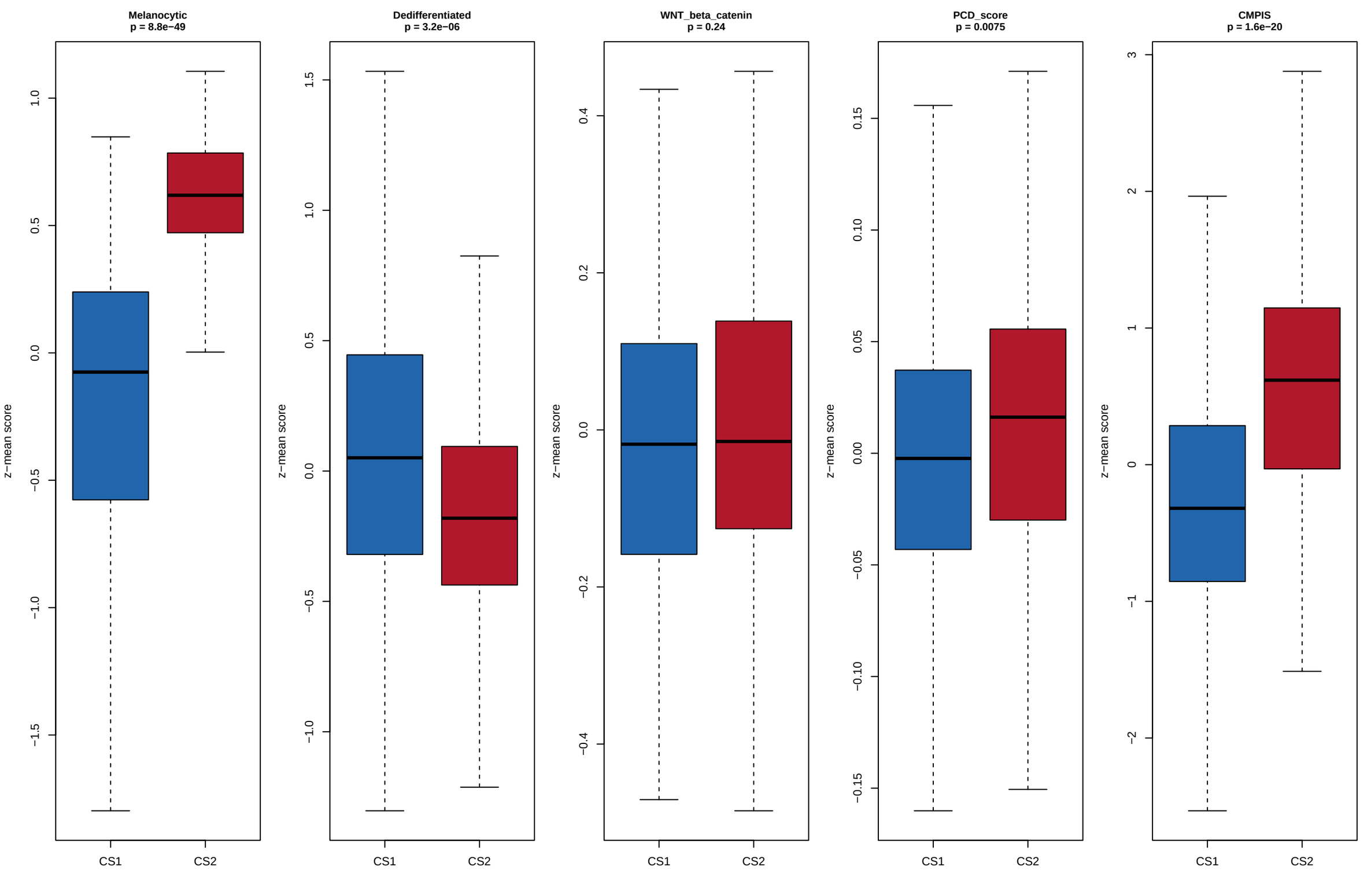

Jerby−Arnon: one point = one tumour (>= 15 malignant cells)
rho = 0.60, p = 0.088 (n = 9)
Dedifferentiated programme
Tumour−intrinsic CS2 − CS1
rho = 0.88, p = 0.0016 (n = 9)
Melanocytic programme
Tumour−intrinsic CS2 − CS1
rho = 0.28, p = 0.46 (n = 9)
CD8 T−cell fraction of non−malignant
Tumour−intrinsic CS2 − CS1
rho = 0.32, p = 0.41 (n = 9)
WNT/beta−catenin score
Tumour−intrinsic CS2 − CS1
rho = 0.22, p = 0.58 (n = 9)
PCD score (malignant)
WNT/beta−catenin score
rho = −0.25, p = 0.52 (n = 9)
CD8 T−cell fraction
CMPIS (malignant mean)


Tirosh
Spearman rho (summed L−R strength vs CS2−CS1)
1.0
0.5
0.0
−0.5
−1.0
malignant −> partner
partner −> malignant
T cell
NK
Macrophage
CAF
Endothelial
B cell


Jerby−Arnon
Spearman rho (summed L−R strength vs CS2−CS1)
1.0
0.5
0.0
−0.5
−1.0
malignant −> partner
partner −> malignant
T cell
NK
Macrophage
CAF
Endothelial
B cell


Tirosh: top 12 L−R pairs (nominal p)
IL10−IL10RA [M> T cell]
LGALS3−LAG3 [M> T cell]
CD86−CD28 [<M T cell]
IFNG−IFNGR2 [M> T cell]
CD86−CD28 [M> T cell]
CXCL13−CXCR5 [<M T cell]
CD80−CTLA4 [M> T cell]
CXCL11−CXCR3 [M> T cell]
CD86−CTLA4 [M> T cell]
PDCD1LG2−PDCD1 [<M T cell]
CD80−CD28 [M> T cell]
CD274−PDCD1 [M> T cell]
0.0
0.2
0.4
0.6
0.8
Spearman rho with tumour CS2 − CS1


Jerby−Arnon: top 12 L−R pairs (nominal p)
CD70−CD27 [M> T cell]
IL10−IL10RA [<M T cell]
MIF−CXCR4 [M> T cell]
ICAM1−ITGAL [M> T cell]
CD80−CTLA4 [<M T cell]
HGF−MET [<M T cell]
CXCL12−CXCR4 [M> T cell]
PROS1−AXL [M> T cell]
HLA−E−KLRD1 [M> T cell]
HGF−MET [M> T cell]
TGFB1−TGFBR1 [M> T cell]
IFNG−IFNGR1 [<M T cell]
−0.4
−0.2
0.0
0.2
0.4
0.6
Spearman rho with tumour CS2 − CS1


Melanocytic
p = 8.8e−49
Dedifferentiated
p = 3.2e−06
WNT_beta_catenin
p = 0.24
PCD_score
p = 0.0075
CMPIS
p = 1.6e−20
z−mean score
CS1
CS2